Chao Zhou (周超)

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Citizenship: Chinese

Research interests

(efficient) machine learning, neural network architecture design, generative model, image and signal processing

Education

University College London (UCL) — London, UK
PhD in electronic and electrical engineering — Sep.2018 – Sep.2023
Supervisor: Professor Miguel Rodrigues

University of Electronic S&T of China (UESTC) — Chengdu, China
Master in Communication and Information System — Sep 2015 – Jun 2018
Mentor: Professor Xuesong Jonathan Tan

Wuhan University of Technology (WHUT) — Wuhan, China
Bachelor in Communication Engineering — Sep 2011 – Jun 2015
Mentor: Professor Kewen Liu

Industry experience

Huawei Technologies Co., Ltd. — Shenzhen, China
Senior R&D Engineer — Jan 2024-Apr 2024

Huawei Technologies Research & Development (UK) Ltd. — London, UK
Research Intern — Jun 2022-Jan 2023

Research experience

Relational Machine Learning Group, CISPA Helmholtz Center for Information Security, Saarbrücken, Germany — **Postdoc**
Supervisor: Dr. Rebekka Burkholz — May.2024 – present
working under sparse-ML Project.

Information, Inference and Machine Learning (I2ML) lab, UCL, London, UK, — **PhD**
Mentor: Professor Miguel Rodrigues — Sep.2018 – Sep.2023
develop algorithms for hyperspectral image unmixing.

Publications

1. **C. Zhou**, T Jacobs, A Gadhikar, R Burkholz. "Pay Attention to Small Weights." accepted by NIPS2025.
2. A Gadhikar, T Jacobs, **C. Zhou**, R Burkholz. (2025). Sign-In to the Lottery: Reparameterizing Sparse Training From Scratch. accepted by NIPS2025.
3. T Jacobs, **C. Zhou**, R Burkholz. "Mirror, Mirror of the Flow: How Does Regularization Shape Implicit Bias?." arXiv preprint arXiv:2504.12883, accepted by ICML 2025.
4. **C. Zhou**, Pu Wei, and M. R. D. Rodrigues. "Neural Network for Blind Unmixing: a novel MatrixConv Unmixing (MCU) Approach." arXiv preprint arXiv:2503.08745 (2025).
5. **C. Zhou** and M. R. D. Rodrigues, "Hyperspectral Blind Unmixing Using a Double Deep Image Prior," in IEEE Transactions on Neural Networks and Learning Systems, doi: 10.1109/TNNLS.2023.3294714.
6. **C. Zhou**, Z. Lyu and M. R. D. Rodrigues, "BITS-Net: Blind Image Transparency Separation Network," 2023 IEEE International Conference on Image Processing (ICIP), Kuala Lumpur, Malaysia, 2023, pp. 375-379, doi: 10.1109/ICIP49359.2023.10222918.
7. W. Pu, B Sober, N. Daly, **C. Zhou**, Z. Sabetsarvestani, C. Higgitt, I. Daubechies, and M. R. D. Rodrigues, "Image Separation with Side Information: A Connected Auto-Encoders Based Approach," in IEEE Transactions on Image Processing, vol. 32, pp. 2931-2946, 2023, doi: 10.1109/TIP.2023.3275872.
8. **C. Zhou** and M. R. D. Rodrigues, "Blind Unmixing Using A Double Deep Image Prior," ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Singapore, Singapore, 2022, pp. 1665- 1669, doi: 10.1109/ICASSP43922.2022.9747545.
9. W. Pu, **C. Zhou**, Y. C. Eldar, and M. R. D. Rodrigues. "Robust lEarned Shrinkage-Thresholding (REST): Robust unrolling for sparse recover." arXiv preprint arXiv:2110.10391 (2021).
10. **C. Zhou** and M. R. D. Rodrigues, "ADMM-Based Hyperspectral Unmixing Networks for Abundance and Endmember Estimation," in IEEE Transactions on Geoscience and Remote Sensing, vol. 60, pp. 1-18, 2022, Art no. 5520018, doi: 10.1109/TGRS.2021.3136336.
11. W. Pu, **C. Zhou**, Y. C. Eldar and M. R. D. Rodrigues, "REST: Robust lEarned Shrinkage-Thresholding Network Taming Inverse Problems with Model Mismatch," ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, ON, Canada, 2021, pp. 2885-2889, doi: 10.1109/ICASSP39728.2021.9414141.

12. **C. Zhou** and M. R. D. Rodrigues, ”An ADMM Based Network for Hyperspectral Unmixing Tasks,” ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, ON, Canada, 2021, pp. 1870-1874, doi: 10.1109/ICASSP39728.2021.9414555.

13. X. J. Tan, **C. Zhou** and J. Chen, ”Symmetric Channel Hopping for Blind Rendezvous in Cognitive Radio Networks Based on Union of Disjoint Difference Sets,” in IEEE Transactions on Vehicular Technology, vol. 66, no. 11, pp. 10233-10248, Nov. 2017, doi: 10.1109/TVT.2017.2726352.

Services

Conference Reviewer: EUSIPCO, ICLR.
Journal Reviewer: IEEE TGRS, IEEE JSTSP, IEEE TIT, IEEE Access.
Area Chairs: CPAL.

Teaching experience

Teaching assistant, UCL Consultants Limited and Turing Innovations Ltd 2020-2023

Machine Learning Masterclass for Defence Science and Technology Laboratory (DSTL), UK

Duties: Mini-project; Answer questions re. mini-project; Weekly programming Assignment/Tasks; Answer questions re. weekly programming assignment/tasks; Answer moodle questions re above; Quizzes; Reading material; Maintain/Fill-in moodle pages.

Post Graduate Teaching Assistant, Dept. of Electronic and Electrical Engineering at UCL Fall 2020 & Fall 2021

ELEC0134: Applied Machine Learning Systems

Duties: Mini-project; Answer questions re. mini-project; Weekly programming Assignment/Tasks; Answer questions re. weekly programming assignment/tasks; Answer moodle questions re above; Quizzes; Reading material; Maintain/Fill-in moodle pages.

Post Graduate Teaching Assistant, Dept. of Electronic and Electrical Engineering at UCL Fall 2019

ELEC0136: Data Acquisition and Processing Systems

The module will cover aspects of data acquisition, data visualization, data cleaning, data analysis and processing.

Honors and scholarships

The First-class National Scholarship of Graduate Students (UESTC) 2017
The Seventh Place of College Fitness of Mianyang Fitness Bodybuilding Championships 2016
The Outstanding Graduate Student (UESTC) 2016
The First-class National Scholarship of Graduate Students (UESTC) 2016

The First-class National Scholarship of Graduate Students (UESTC) 2015

Advanced Individuals of Innovation and Entrepreneurship for work of Communist Youth League (WHUT) 2014

The National Endeavor Fellowship of Undergraduate Students, (WHUT) 2013

The First-Class Prize of National Undergraduate Electronics Design Contest (WHUT) 2013

The University Outstanding Student Leaders (WHUT) 2013

The Second-Class Prize of the Sixth Electric and Electronic Innovation Design Contest (WHUT) 2013

The School Outstanding Student (WHUT) 2012

Skills

Programming

Proficient in: python, pytorch, matlab.

Familiar with: C, Java.

Languages

Chinese, English

Other interests

Bodybuilding.